SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 3

ERROR ANALYSIS TASK

COURSE NAME: DIGITAL SIGNAL PROCESSIN G COURSE CODE: ECA09

QUESTION:

3. An 8-bit ADC with ±5 V range reports a quantization noise power of 0.01 V^2. Verify the calculation and identify the error.

INTRODUCTION:

An Analog-to-Digital Converter (ADC) converts an analog signal into a digital signal. During quantization, a small error called quantization error is produced. This problem verifies the quantization noise power of an 8-bit ADC and identifies the calculation error.

Given Data

- Resolution, bits

- Input Voltage Range = to
- Reported Quantization Noise Power =

Step 1: Calculate the FullScale Range

FORMULA :

-

Substitute the Values

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-
-

Step 2: Calculate the Quantization Step Size (Q)

- **Formula**
-
- **Substitute the Values**
-
- **Therefore,**
-
- ## Step 3: Calculate the Variance of Quantization Error
- **Formula**

-

- **Substitute the Values**

-

- **Therefore,**

-

Step 4: Calculate the Quantization Noise Power

- **Formula**

-

- **Substitute the Values**

-

- **Therefore,**

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- ## Step 5: Verification

- **Reported Quantization Noise Power**

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- **Calculated Quantization Noise Power**

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- Since,
-

- The reported value is **incorrect**.

- ## Step 6: Error Identification
- The reported value is incorrect because:
- The quantization step size ( ) may have been calculated incorrectly.
- The variance formula may have been applied incorrectly.
- A decimal-point or arithmetic error may have occurred.
- An incorrect full-scale range may have been used.

- ## Result
- **Parameter** **Value**
- Full Scale Range (R) **10 V**
- Resolution (b) **8 bits**
- Quantization Step Size (Q) **0.0390**

- Variance of Quantization Error ( ) **6 V**
- Reported Value **0.01 V²**
- Verification **Incorrect**

- ## Conclusion

- The **quantization step size** of the 8-bit ADC is **0.03906 V**, and the **quantization noise power** is . Therefore, the reported value of **0.01 V²** is incorrect because it is much higher than the calculated theoretical value. The error is likely due to incorrect formula substitution or calculation mistakes.